3 Fig. 3 shows a circle with centre O and radius 1 unit. Points A and B lie on the circle with angle AOB = θ radians. C lies on AO, and BC is perpendicular to AO.

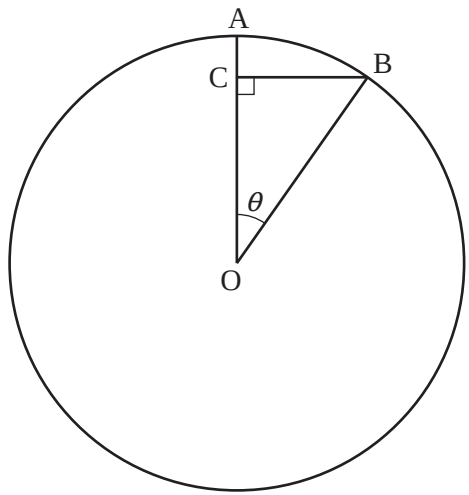


Fig. 3

Show that, when θ is small, $AC \approx \frac{1}{2}\theta^2$. **[2]**

3